

**BUDDY: An Intelligent Robot Dog for Offline Voice
Interaction, Visual Perception, and Safety-Aware
Multimodal Control**

INDIVIDUAL THESIS

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Individual component: Offline Voice Intelligence, Personalized
Interaction, and Safety-Aware System Integration for an Intelligent
Companion Robot

Group ID: R26-SE-040

Report submitted in partial fulfilment of the requirements for the
BSc (Hons) in Information Technology Specialising in Software Engineering

Specialisation: Software Engineering

Department of Computing

Sri Lanka Institute of Information Technology

Sri Lanka

October 2026

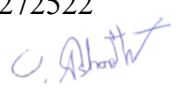
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ABSTRACT

This thesis presents the individual contribution to BUDDY focused on offline voice intelligence, personalized interaction, and safety-aware system integration for a low-cost companion robot dog. The work addresses a central embodied-AI problem: spoken or conversational output should not obtain unrestricted authority over physical actuators. A local auditory pipeline was therefore developed around an AB13X USB microphone, 48 kHz stereo capture converted to 16 kHz mono, Vosk offline speech recognition, a dedicated wake-word state, deterministic movement grammar, a conversational response layer with persistent user memory and machine-learning intent fallback, and Piper text-to-speech. Physical movement requests are published through ROS 2 as structured MovementCommand messages and must pass through an independent safety supervisor before reaching the motor executor. The implemented prototype was validated on Raspberry Pi 4 hardware. Cold-boot operation reached an interactive state without manual terminal launch; the wake phrase remained usable from 0-3 m in the tested environment and became inconsistent around 3.4 m; spoken forward, backward, left, right, and stop commands produced the expected physical wheel behaviour; and an active forward request was interrupted when a front obstacle was introduced and resumed after obstacle removal. Four-direction ultrasonic sensing and the safety/motor software tests also operated successfully. Dual-identical-USB-microphone acoustic localization remained experimental because reliable simultaneous capture was not achieved. The work demonstrates that practical offline speech interaction can be combined with modular ROS 2 control while preserving an explicit safety boundary between conversational intelligence and physical actuation.

Keywords: offline speech recognition, human-robot interaction, ROS 2, safety-aware control, companion robot

ACKNOWLEDGEMENTS

I would like to express my sincere gratitude to my supervisor, Pradeep K. W. Abeygunawardhana, and co-supervisor, Samantha Rajapakse, for their guidance throughout this research. I also thank my research teammates, Sobiya Anton Suresh and S. S. Y. Wickramasinghe, for their collaboration in integrating visual intelligence, navigation, sensing, and application-level functionality with the BUDDY platform. Their subsystem contributions made it possible to evaluate the voice component within an embodied multimodal robot rather than as an isolated software demonstration.

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LIST OF ABBREVIATIONS

Abbreviation	Meaning
ASR	Automatic Speech Recognition
HRI	Human–Robot Interaction
ROS 2	Robot Operating System 2

1 INTRODUCTION

1.1 Overall project context

Companion robots combine sensing, interaction, mobility and physical embodiment in a shared human environment. Unlike a conventional software assistant, an embodied system must interpret user input while simultaneously respecting mechanical, electrical and environmental constraints. BUDDY was developed as an affordable intelligent robot-dog prototype that integrates local speech interaction, visual perception, environmental sensing, four-wheel mobility and extensible companion behaviours on Raspberry Pi-class hardware. The broader project is motivated by the gap between feature-rich commercial robotic companions and low-cost platforms that are sufficiently open for research, modification and privacy-conscious local processing [1], [2].

The complete BUDDY system contains three principal individual research streams: offline auditory interaction and system integration; visual and facial intelligence; and navigation, environmental perception and autonomous behaviour. These streams are joined through a modular ROS 2 architecture in which high-level components publish requests rather than directly commanding physical hardware. The most important systems constraint is that uncertain conversational or AI-generated output must not bypass safety supervision when physical motion is involved. The group paper therefore positions the project around multimodal integration and explicit separation between high-level movement intent and motor authority.

The candidate's contribution focuses on the auditory and integration path. This includes microphone acquisition, local speech recognition, wake/sleep interaction state, deterministic movement command handling, conversational response logic, persistent user information, local text-to-speech, ROS 2 movement-message integration, automatic boot deployment, hardware/software integration and participation in end-to-end physical validation. The contribution is not presented as a new speech-recognition algorithm; rather, the research investigates how established speech and robot middleware techniques can be combined into a practical, low-cost, safety-aware embodied interaction architecture.

1.2 Individual component context

Voice is a natural interface for a companion robot because it allows hands-free interaction and can support both social conversation and physical commands. However, these two classes of interaction have different risk profiles. A conversational reply may tolerate ambiguity, but a physical command such as “move forward” cannot safely be treated as ordinary generated text. A mistaken or stale interpretation may cause a collision, and the environment may change after a command is accepted. The BUDDY voice component was therefore designed around two complementary principles: local-first interaction for basic availability and privacy, and deterministic separation of safety-relevant commands from unconstrained conversational intelligence.

The physical runtime uses an AB13X USB microphone. Audio is captured as 48 kHz, two-channel PCM, the channels are combined, and the signal is converted to 16 kHz mono for Vosk recognition. Separate wake, command and general recognizers constrain recognition differently according to robot state. The wake phrase “Buddy” activates the robot, deterministic movement phrases are mapped to structured MovementCommand messages, and general utterances are passed to a conversational response layer. Spoken output is generated locally with Piper. This organization allows the system to remain useful without requiring a cloud speech service for core interaction.

1.3 Problem statement

The individual research problem is how to provide practical offline voice interaction on resource-constrained companion-robot hardware while preventing natural-language processing from obtaining unrestricted physical actuation authority. The challenge is broader than transcription accuracy. The subsystem must manage microphone constraints, wake-word state, command ambiguity, response latency, persistent interaction context, ROS 2 communication, sensor-driven safety intervention, and embedded deployment. It must also degrade safely when recognition or environmental information is uncertain.

A second problem concerns integration. Voice software that works on a laptop is insufficient evidence for an embodied robot. The auditory path must coexist with ultrasonic sensing, motor execution, visual processes, system services and physical power domains. A valid contribution therefore requires evidence that the voice path operates through the real ROS 2 control architecture on the Raspberry Pi and that safety remains independent from the recognized command source.

1.4 Research gap

Modern cloud speech systems achieve strong recognition quality, but continuous cloud dependence can add network dependency, privacy exposure and service availability constraints. Fully local systems avoid these issues but operate under tighter computational and acoustic limitations. Research in socially assistive robotics also shows that useful autonomy depends on integration between perception, interaction and actuation rather than isolated AI modules [2]. ROS 2 provides a modular basis for such integration [4], yet application-level designs still determine whether high-level intelligence is permitted to directly influence hardware.

The practical gap addressed here is therefore not the absence of speech recognizers. It is the need for a low-cost implementation pattern in which local speech can control a real companion robot while remaining explicitly subordinate to an independent environmental safety layer. BUDDY contributes an implemented case study of this separation and evaluates the complete path from local audio acquisition to safety-mediated physical wheel motion.

1.5 Aim and objectives

The aim of this individual research is to design, implement and validate an offline-first voice-intelligence subsystem and its safety-aware integration with the BUDDY companion robot.

Objective	Description
O1	Implement local wake-word, speech-recognition and text-to-speech interaction suitable for Raspberry Pi-class hardware.
O2	Separate safety-critical movement grammar from general conversational processing.
O3	Publish recognized physical commands as structured ROS 2 movement requests rather than direct motor operations.
O4	Integrate the voice subsystem with an independent obstacle-aware safety supervisor and motor executor.
O5	Provide persistent user memory and a lightweight machine-learning fallback for non-safety conversational intents.
O6	Deploy the integrated runtime so that BUDDY can start automatically after power-up.
O7	Evaluate the physical voice-to-motion path, wake range, safety intervention and key software behaviours without inventing unsupported accuracy statistics.

1.6 Research questions

RQ1: Can offline speech interaction provide practical voice-controlled operation of a low-cost companion robot on Raspberry Pi-class embedded hardware?

RQ2: Can an independent environmental safety layer prevent unsafe continuation of otherwise valid voice-generated movement requests?

RQ3: How effectively can auditory interaction be integrated with ROS 2, persistent user context, environmental sensing and physical actuation while preserving clear authority boundaries?

1.7 Scope limitations and assumptions

The validated scope is an indoor prototype operating with a single primary USB microphone and a constrained English movement vocabulary. The implemented physical command set includes forward, backward, left, right and stop. General conversation is intentionally separated from physical motion. The system is evaluated as a functional research prototype rather than a production-certified personal-care robot.

A controlled multi-participant speech-recognition experiment was not completed, so generalized command accuracy or word-error rate is not reported. Two identical AB13X USB microphones were investigated for acoustic localization, but reliable simultaneous capture was not achieved on the Raspberry Pi USB configuration. Sound-direction localization therefore remains experimental and is not presented as a validated result. Internet-assisted capabilities and local LLM experiments were developed in earlier software prototypes but are not treated as part of the physically validated safety-critical runtime.

1.8 Individual contribution

The candidate designed and implemented the core offline auditory pipeline, conversational brain and ROS 2 voice integration, and led substantial system-integration work required to move from separate modules to a functioning physical robot. Repository artefacts include the voice node, conversational brain, persistent user-profile support, machine-learning intent model, deployment scripts and service configuration. The candidate also contributed to motor/sensor/neck integration and final troubleshooting so that the voice path could be validated against real physical behaviour.

Where the candidate assisted other members, ownership is kept explicit. For example, face/emotion intelligence remains Sobiya Anton Suresh's research component even where the candidate assisted with final physical camera-head integration or dependency troubleshooting. Navigation and autonomous planning remain S. S. Y. Wickramasinghe's research component even though the voice and safety integration uses environmental outputs from that subsystem.

1.9 Thesis structure

Chapter 2 reviews relevant work in companion robotics, offline speech processing, human-robot interaction, ROS 2 and safety-oriented control. Chapter 3 describes the research methodology and evidence rules. Chapter 4 presents the product and component design. Chapter 5 details implementation, integration, deployment and testing. Chapter 6 reports the validated results. Chapter 7 interprets the findings and limitations. Chapter 8 considers commercialisation and sustainability. Chapter 9 concludes the thesis and identifies future work.

2 LITERATURE REVIEW AND RELATED WORK

2.1 Review method and scope

The literature review was scoped around technologies and research questions directly relevant to the individual component: companion and socially assistive robotics, local speech recognition, wake-word style interaction, lightweight intent classification, robot middleware, and safety separation between high-level intelligence and physical actuation. Priority was given to peer-reviewed sources that establish the research foundations, while implementation documentation was used for software packages such as Vosk, Piper and ROS 2 where appropriate.

The review intentionally distinguishes algorithmic novelty from system integration. BUDDY does not claim to introduce a new acoustic model or a novel text classifier. The contribution is evaluated as an embodied systems design in which speech, user context and safety constraints are coordinated on affordable hardware.

2.2 Domain and theoretical foundations

Companion robots are expected to combine physical embodiment with sustained human interaction. Reviews of human-robot companionship identify zoomorphic forms, social engagement and mobility as important design directions [1]. Socially assistive robotics similarly emphasizes autonomy, sensing, user interaction and physical action as interdependent capabilities [2]. This is particularly relevant to BUDDY because spoken commands influence a moving platform rather than a screen-only assistant.

Automatic speech recognition converts audio into textual hypotheses. Large-scale systems such as Whisper show the potential of data-intensive speech models [3], while Vosk builds on Kaldi-style decoding to provide lightweight offline recognition suitable for embedded and desktop use [5], [6]. For a constrained robot command vocabulary, grammar restriction can reduce the space of possible outputs and simplify deterministic mapping from recognized phrases to robot requests.

Text-to-speech closes the interaction loop. BUDDY uses Piper for local neural speech synthesis in the physical runtime. Keeping basic speech synthesis local avoids

dependence on cloud availability and reduces the need to transmit user speech or generated responses to external services. The resulting local-first architecture is consistent with the project's privacy and availability goals.

2.3 Related research

Speech interfaces have been widely used in service and social robots, but a key systems issue is how recognized speech is connected to physical action. In many demonstrations the recognition stage and motor stage are presented sequentially, while less attention is given to authority separation when the environment changes after command recognition. Behaviour-based robotics research established the value of layered control in which lower-level safety behaviours can override higher-level goals [9]. BUDDY applies a related principle through an explicit ROS 2 safety supervisor between movement requests and the motor executor.

ROS 2 was selected because it supports modular nodes, typed interfaces and distributed communication patterns appropriate to multi-component robots [4]. The `MovementCommand` message is used as a request structure rather than a direct GPIO instruction. This allows voice, navigation or other future command sources to share the same safety boundary instead of implementing independent motor logic.

For non-safety conversation, BUDDY combines deterministic rules with a lightweight TF-IDF and logistic-regression classifier. Scikit-learn provides mature implementations of these techniques [8]. The role of this classifier is not to authorize movement but to improve recognition of conversational intent such as greeting, time/date requests, wellbeing, thanks and name-related interactions.

2.4 Existing systems, products and techniques

Commercial robotic pets and research platforms often provide sophisticated perception or locomotion but may be expensive, closed, or dependent on proprietary ecosystems. At the opposite end, hobby robots may provide inexpensive movement but limited natural interaction and safety architecture. BUDDY targets the middle ground: a reproducible research platform using replaceable commodity components, open software tools and explicit subsystem boundaries.

Technique / system	Strength	Limitation relevant to BUDDY
Cloud speech assistant	High-quality recognition and broad language capability	Depends on connectivity and external services for core interaction.
Offline ASR with Vosk/Kaldi	Local processing, predictable availability, small models	Recognition can degrade with noise, distance and accent; constrained hardware limits model size.
Direct command-to-motor scripting	Simple to implement	High-level recognition can gain unsafe physical authority.
ROS 2 modular control	Typed interfaces, process separation, reusable nodes	Safety still depends on application architecture and topic authority.
BUDDY approach	Offline voice + structured movement request + independent safety supervisor	Prototype-scale validation; not a certified safety system.

2.5 Gap synthesis and conceptual framework

The literature supports the individual building blocks used by BUDDY, but the research gap lies in their low-cost embodied integration. The conceptual framework for this thesis separates interaction into three levels: perception, intention and authorization. Audio perception produces recognized text; the voice subsystem converts relevant text into an intention such as a MovementCommand; and the safety subsystem independently decides whether physical authorization should be granted. This separation is the basis for the design and evaluation that follow.

3 RESEARCH METHODOLOGY

3.1 Research design

A design-science and iterative prototyping methodology was used. The artifact under study is the BUDDY offline voice and safety-aware integration subsystem. Development progressed from software-only voice experiments, through ROS 2 message integration, to physical deployment on Raspberry Pi hardware with motors, ultrasonic sensors and speaker output. Each iteration was accepted only when its observable behaviour could be reproduced without contradicting the project's safety constraints.

The evaluation combines component-level verification with end-to-end physical experiments. Software tests provide evidence that message handling, configuration, hardware abstraction and safety logic behave as intended under controlled inputs. Physical tests establish that the complete chain also produces the expected behaviour on the real robot. These forms of evidence are reported separately because passing software tests does not establish physical reliability under every environmental condition.

3.2 Data, participants and study materials

The principal study materials were the BUDDY hardware prototype, Raspberry Pi 4B with 4 GB RAM, Ubuntu 24.04 LTS ARM64, ROS 2 Jazzy, AB13X USB microphone, local speaker output, four HC-SR04 ultrasonic sensors, two TB6612FNG motor drivers and four DC geared motors. The validated voice experiments used the candidate as the primary speaker in a normal indoor environment. A multi-speaker controlled trial was not completed and no generalized population-level speech metric is claimed.

Software materials included the Vosk small English model, Piper en_US-lessac-medium voice model, custom ROS 2 interfaces, YAML configuration, Python voice modules, scikit-learn intent classifier, deployment service files and automated software tests. Repository history and physical test notes were used as reproducibility artefacts.

3.3 Experimental and study procedure

The main evaluation procedure contained four stages. First, the microphone and local recognition path were verified independently. Second, recognized movement phrases were mapped to MovementCommand requests and observed on ROS 2 topics without energizing physical motors. Third, the motor and ultrasonic subsystems were enabled through the hardware abstraction and safety supervisor. Finally, cold-boot and obstacle-intervention tests were performed on the complete prototype.

1. Cold boot: disconnect and restore Raspberry Pi power without manually opening an SSH session.
2. Confirm automatic runtime startup and the spoken readiness message.
3. Present the wake phrase “Buddy” and confirm the “Yes?” response.
4. Issue forward, backward, left, right and stop commands and observe physical wheel behaviour.
5. While a forward request remains active, introduce an obstacle into the front ultrasonic region and confirm that the safety layer stops motion.
6. Remove the obstacle and observe whether the still-active request can be reconsidered and motion resumed.
7. Repeat software-level safety and motor tests to confirm that the observed physical behaviour is consistent with expected control logic.

3.4 Measures and evaluation metrics

The evaluation emphasizes directly observable functional measures because a statistically controlled speech corpus study was outside the completed scope. Wake usability was recorded across tested distances. Movement commands were scored by whether the intended physical behaviour occurred through the complete safety-controlled path. Obstacle intervention was evaluated by whether an active command was stopped when the environment changed and could resume when the hazard was removed.

Software verification used pass/fail outcomes from the implemented test suite. Additional timing observations were retained where actually measured, including persistent Piper model load and short-utterance synthesis time. No unsupported

word-error rate, confusion matrix, false-accept rate or generalized latency figure is introduced.

3.5 Data analysis

Results were analysed descriptively. The purpose was to determine prototype feasibility and identify integration boundaries rather than estimate population parameters. Functional observations were compared with the stated objectives and the safety architecture. Failures and partial results, particularly the dual-microphone localization experiment, were retained as limitations rather than removed from the narrative.

3.6 Ethics, privacy, security and responsible AI

Offline recognition reduces the need to transmit core spoken interaction to external cloud services. Persistent memory is stored locally and is limited to interaction context such as a preferred name and recent conversation fields. The design does not claim that local processing alone guarantees security; filesystem access, application permissions and remote interfaces must still be protected in a production system.

Responsible AI is especially important where language interacts with physical movement. The architecture therefore prohibits general conversational output from directly controlling motors. Movement is represented by a small deterministic command vocabulary and remains subject to current environmental state. The implementation is a research prototype and is not presented as compliant with formal personal-care robot certification, though relevant safety principles were considered [10].

3.7 Reproducibility and research artefacts

The project repository contains the ROS 2 workspace, voice node, conversational brain, intent model, interface definitions, hardware abstraction, sensor and motor nodes, safety supervisor, YAML configuration, deployment script and systemd unit. A known working physical-demo commit is 9a3ae883 (“feat: add autonomous boot and voice-controlled physical demo”). The principal runtime path is documented in `scripts/start_buddy_demo.sh` and `deployment/buddy.service`.

4 PRODUCT AND SOLUTION DESIGN

4.1 Requirements

ID	Requirement	Priority
R1	Core wake-word and movement interaction shall operate without a cloud speech service.	High
R2	General conversational logic shall not directly access motor GPIO or motor-driver APIs.	Critical
R3	Movement requests shall pass through the independent safety supervisor.	Critical
R4	The system shall support explicit STOP at all relevant interaction states.	Critical
R5	The voice system shall recover from its own speech output without treating that output as user input.	High
R6	The robot shall return to a safe sleeping state after prolonged inactivity.	Medium
R7	The runtime shall support automatic startup after embedded-system boot.	High
R8	Conversational interactions may use local persistent memory and ML fallback, but physical commands shall remain deterministic.	High
R9	The system shall support modular future integration with visual emotion and navigation components.	Medium

4.2 Component architecture within the complete solution

The voice subsystem is divided into acquisition, recognition, interaction logic, speech synthesis and motion-request publication. Audio is captured from the AB13X microphone, normalized for the recognition engine, and processed by state-specific Vosk recognizers. Movement phrases enter a deterministic mapping path. Other

recognized text is handled by the conversational brain. Both branches provide user feedback through local speech synthesis, but only the movement branch publishes MovementCommand requests.

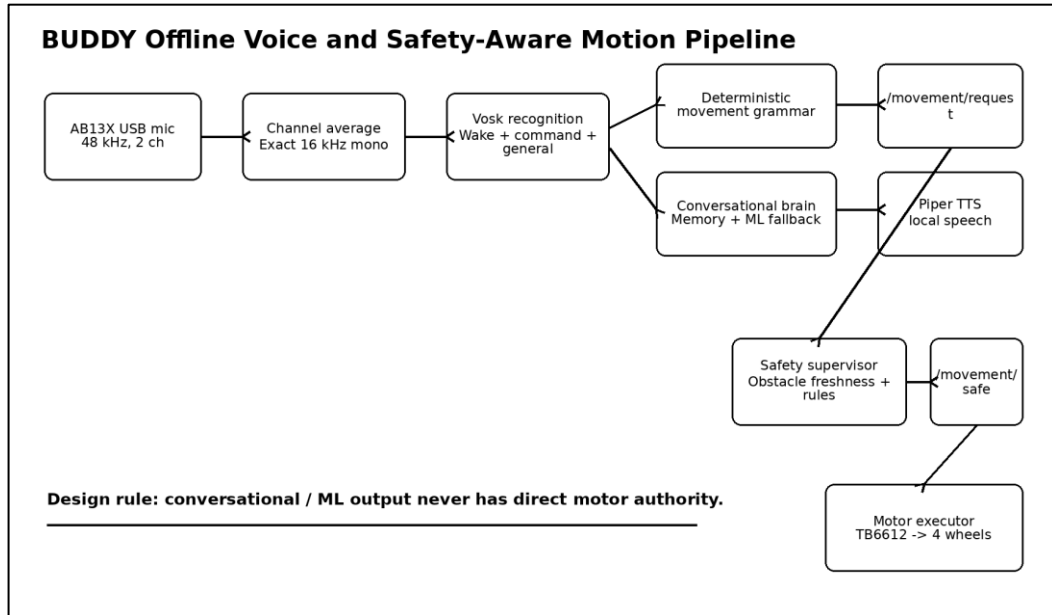


Figure 4.1. BUDDY offline voice and safety-aware motion pipeline.

This design prevents the conversational brain, machine-learning intent classifier and any optional local LLM experiment from becoming a physical motor controller. The safety supervisor consumes obstacle status separately and publishes only safety-approved movement to the motor executor.

4.3 Design alternatives and decisions

Cloud ASR was rejected for safety-critical core commands because network availability should not determine whether the robot can understand STOP or basic movement requests. Vosk was selected for the physical runtime because it operates locally and supports constrained grammars. A larger local model could improve general transcription but would increase resource demand. The design therefore uses a small model plus separate recognition modes rather than applying one unrestricted recognizer to every state.

Direct function calls from the voice script to motor GPIO were also rejected. Although simpler, that approach would couple uncertain audio interpretation with actuator authority and duplicate safety logic across command sources. Publishing a

typed MovementCommand request allows the safety supervisor to remain the unique authorization boundary.

For conversational intelligence, a hybrid approach was selected. Frequently used interactions are handled deterministically; a TF-IDF and logistic-regression classifier provides a lightweight fallback for non-safety intents; and earlier experiments explored optional local LLM support. The physically validated runtime deliberately keeps this intelligence outside the safety-critical movement mapping.

4.4 Detailed design

4.4.1 Audio preprocessing and recognizers

The AB13X device is detected automatically by name and input-channel capability. The microphone provides two input channels at 48 kHz. The voice path combines the channels and reduces the sampling rate to the 16 kHz rate expected by the Vosk model. Wake and movement recognizers use restricted grammar lists, while a separate general recognizer handles conversational text. This reduces cross-contamination between broad natural-language recognition and safety-relevant command interpretation.

4.4.2 Wake and interaction state

BUDDY starts in a sleeping state with the wake recognizer active. Recognizing “Buddy”, “hey buddy”, “hello buddy” or related variants transitions the node to the awake state and produces “Yes?”. Stop is deliberately available even when the system is otherwise sleeping. After approximately 60 s of inactivity, the node publishes STOP, clears queued audio, resets recognizers and returns to sleeping.

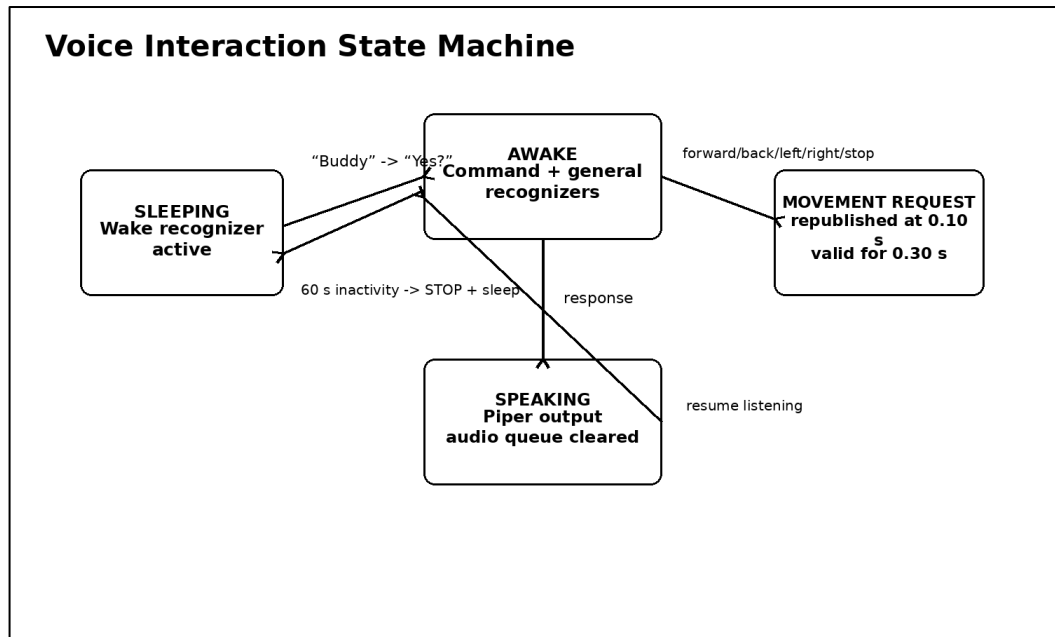


Figure 4.2. Voice interaction state machine used by the physical runtime.

4.4.3 Deterministic physical command mapping

Movement phrases are normalized to a small set of commands: forward, backward, rotate left, rotate right and stop. Forward/backward requests use a normalized speed of approximately 0.20 in the physical demo, while pivot turns use approximately 0.60 because lower values did not produce reliable platform rotation. These values are prototype calibration settings rather than universal motion parameters.

4.4.4 Conversational brain and persistent context

The conversational layer stores the preferred user name, interaction count and recent conversation fields in local JSON data. Deterministic responses cover greetings, identity, time/date, wellbeing, thanks, apology, jokes and capability questions. A scikit-learn classifier uses TF-IDF unigram/bigram features with logistic regression for fallback intent classification. The trained model is loaded from model.pkl and vectorizer.pkl; low-confidence predictions are treated as unknown rather than forcing a response.

4.4.5 Speech output and echo control

Piper provides local neural text-to-speech in the integrated ROS 2 runtime. During speech, the node marks itself as speaking, writes a temporary WAV file, plays it through the configured USB audio device, then clears queued microphone samples

and resets recognizers. This reduces the risk that BUDDY interprets its own synthesized speech as a new user command.

4.4.6 Safety authorization

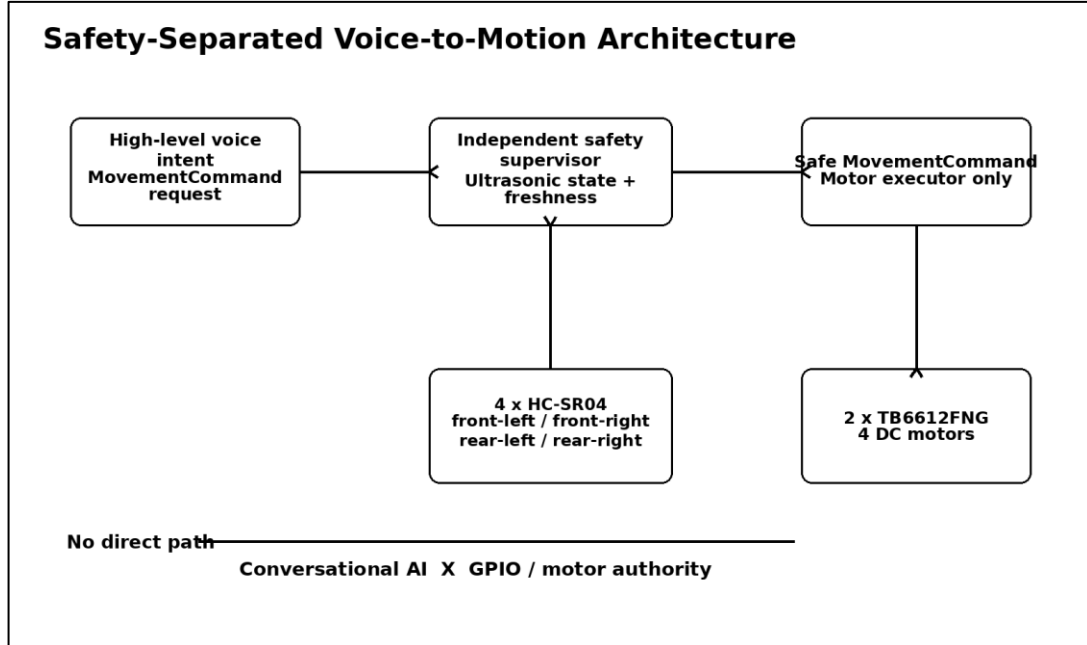


Figure 4.3. Safety-separated voice-to-motion control architecture.

The voice node publishes to `/movement/request`. The independent safety supervisor combines the requested direction with front/rear obstacle status, freshness and emergency state, then publishes allowed commands to `/movement/safe`. The motor executor subscribes only to this safety-approved topic. Environmental observations older than approximately 1.0 s are considered stale during active motion; safety evaluation occurs at approximately 0.1 s intervals, and the motor layer contains an additional approximately 0.05 s watchdog.

4.5 Standards and professional practice

Professional practice considerations include fail-safe defaults, constrained authority, separation of power domains, documented GPIO mappings, version-controlled configuration and explicit distinction between experimental and validated capabilities. The project does not claim formal safety certification. ISO 13482 principles relating to personal-care robot risk awareness were treated as contextual guidance rather than a compliance claim [10].

5 IMPLEMENTATION, INTEGRATION AND TESTING

5.1 Development environment and technology stack

Layer	Technology / configuration
Embedded computer	Raspberry Pi 4 Model B Rev 1.2, BCM2711, 4 GB RAM
Operating system	Ubuntu 24.04 LTS ARM64
Robot middleware	ROS 2 Jazzy
Programming	Python; shell; YAML; ROS 2 interface definitions
ASR	Vosk / KaldiRecognizer, small English model
TTS	Piper en_US-lessac-medium.onnx
Intent ML	scikit-learn TF-IDF + LogisticRegression
Audio input	AB13X USB Audio, S16_LE, 2 channels, 48 kHz
Motion	2 x TB6612FNG, 4 DC geared motors
Environmental sensing	4 x HC-SR04 via level conversion
Deployment	systemd service + start_buddy_demo.sh

5.2 Component implementation

5.2.1 ROS 2 voice node

BuddyVoiceNode creates a publisher for MovementCommand on /movement/request and maintains an active command that is republished every 0.10 s while movement remains requested. Each message includes timestamp, sequence ID, source identifier, command, speed values, short validity duration and emergency flag. Repeated short-

lived requests allow the downstream safety layer to reason about freshness rather than relying on one indefinite command.

5.2.2 Recognition pipeline

The physical implementation uses separate KaldiRecognizer instances for wake, constrained command and general speech. While sleeping, only wake/stop phrases are relevant. While awake, the command recognizer and general recognizer process the same normalized audio stream, but movement phrases receive priority. A short guard interval prevents the same utterance from being immediately handled again as general text after a movement command.

5.2.3 Intent model

The non-safety intent classifier is trained from intents.json. Training uses TfidfVectorizer with unigram and bigram features, sublinear term frequency and logistic regression with $C=2.0$ and a 500-iteration maximum. Cross-validation is printed when class counts permit it, but the thesis does not report an accuracy value unless reproduced from a controlled saved run. In the deployed conversational brain the ML classifier is a fallback behind deterministic rules.

5.2.4 Persistent memory

A local memory file allows BUDDY to retain a preferred name and basic conversation continuity across process restarts. The design intentionally limits persistence to small interaction fields rather than recording raw continuous microphone audio. A separate user-profile helper stores robot/user preferences and notes for future personalization research.

5.2.5 Local speech synthesis

Piper is loaded once and retained for repeated synthesis. Earlier timing observations showed an initial model load of approximately 5.776 s and short utterance synthesis around 0.24-0.40 s after loading. These observations are useful for understanding startup cost versus steady-state responsiveness; they are not treated as a comprehensive latency benchmark.

5.2.6 Extended conversational and utility prototype

The candidate also developed a broader software prototype outside the minimal physical-demo path. This branch explored automatic English/Tamil text-language detection, selectable voice profiles such as calm, energetic, whisper and normal modes, lightweight text-emotion cues, persistent personality/topic learning, and optional local Llama 3.2 responses through Ollama. Additional utility functions included music playback through yt-dlp/VLC, weather and web-summary requests, calculator-style expressions and distance calculations. These capabilities demonstrate the intended extensibility of BUDDY as a companion assistant, but they are reported as software-level or optional features rather than physically validated safety-critical functions.

The design boundary is important: even when optional conversational or internet-assisted intelligence is enabled, it does not obtain direct GPIO or motor authority. Physical motion remains restricted to deterministic movement intents and the ROS 2 safety-controlled path. This allows richer interaction to evolve independently from the validated movement-safety architecture.

5.3 Integration with the other components

The candidate integrated the voice subsystem with BUDDY's shared ROS 2 interfaces and physical control chain. Four ultrasonic sensors provide front-left, front-right, rear-left and rear-right observations. The safety supervisor evaluates those observations before movement reaches the motor executor. This allows the same safety mechanism to mediate voice, future navigation requests and other command sources.

The candidate also contributed to overall hardware and subsystem integration: Raspberry Pi configuration, motor-driver mappings, ultrasonic GPIO and level shifting, PCA9685/servo setup, power-domain separation, common grounding, system startup and troubleshooting. Assistance was also provided during visual subsystem integration, including camera-head and dependency issues, but the underlying face/emotion research remains assigned to Sobiya Anton Suresh.

The current system separates the Raspberry Pi compute supply from the higher-current motor/servo rail. The Pi is powered through a dedicated USB-C power bank, while the actuation rail is derived from a protected 3S 18650 battery arrangement through regulation to approximately 4.996 V. This separation reduced the risk that transient actuator loads destabilize the computing platform.

5.4 Verification and software testing

The integrated repository includes tests for configuration, hardware abstraction, logging, motors, neck control, safety supervision and secrets. Thirteen relevant software tests were reported as passing during development. These tests support confidence in expected control logic but are intentionally reported separately from physical validation.

Test area	Purpose	Status
Config / secrets	Verify configuration loading and safe handling of environment-specific settings	Passed in reported suite
Hardware abstraction	Verify mock/physical backend behaviour and safe initialization	Passed in reported suite
Motor control	Verify movement mapping and driver-level control logic	Passed in reported suite
Safety supervisor	Verify command gating, obstacle logic and stale-state handling	Passed in reported suite
Neck controller	Verify bounded pan/tilt control logic	Passed in reported suite
Integrated relevant tests	Combined software verification reported in group evaluation	13 / 13 passed

5.5 Deployment and operation

Embedded deployment is implemented through deployment/buddy.service, which launches scripts/start_buddy_demo.sh after operating-system startup. The script

sources ROS 2 and the workspace, activates the Python environment, then starts the safety supervisor, physical ultrasonic sensor node, physical motor executor and voice runtime with controlled delays and log redirection. Process IDs are retained so that shutdown cleanup can terminate child processes.

A cold-boot experiment demonstrated that BUDDY could reach the interactive voice state without manually launching a terminal or SSH session. After power restoration the robot produced “Hi there. Buddy is ready.”, responded to “Buddy” with “Yes?”, and accepted physical movement commands after the motor battery was enabled. This result is important because it demonstrates an embedded robotic appliance workflow rather than a developer-only script demonstration.

6 RESULTS AND VALIDATION

6.1 Research results

The completed experiments demonstrate the feasibility of offline speech-controlled operation through an explicit safety boundary. The strongest result is not recognition in isolation; it is that recognized movement remained a request while environmental state independently retained the ability to stop or permit physical movement. This validates the central architectural principle of the individual contribution.

6.2 Product and technical results

Experiment	Condition	Observed result	Evidence level
Automatic startup	Cold boot; no manual terminal launch	BUDDY runtime started and produced readiness speech	Physically validated
Wake interaction	0-3.0 m	Wake-word interaction remained usable in the tested environment	Physically validated
Extended wake range	Approx. 3.4 m	Recognition became inconsistent	Physical observation
Forward	Spoken command	Robot moved forward through safety-controlled path	Physically validated
Backward	Spoken command	Robot moved backward	Physically validated
Left	Spoken command	Robot performed left pivot/turn	Physically validated
Right	Spoken command	Robot performed right pivot/turn	Physically validated
Stop	Spoken command	Active movement stopped	Physically validated

Four-direction sensing	FL, FR, RL, RR channels	All four ultrasonic channels produced usable observations	Physically validated
Dynamic obstacle stop	Obstacle introduced during active forward request	Safety layer stopped forward movement	Physically validated
Obstacle removal	Obstacle removed while request remained active	Request was reconsidered and movement resumed	Physically validated
Software verification	Relevant test suite	13 tests completed successfully	Software-level
Persistent memory	Name / conversation fields	State retained locally across interactions	Software-level
Dual microphone localization	Two identical AB13X devices	Reliable simultaneous capture not established	Experimental / not validated

The wake-distance observation is not a generalized speech-recognition range specification. It records behaviour in the tested physical environment with the current microphone placement and speaker. Similarly, the movement results demonstrate end-to-end functional behaviour rather than command-recognition accuracy across different speakers, accents and noise levels.

6.3 User or stakeholder validation

The prototype was demonstrated as a natural hands-free interface in which the user can wake BUDDY, receive spoken acknowledgement and issue movement instructions without a keyboard or remote control. The interaction pattern was suitable for the project's companion-robot concept. Formal usability questionnaires

or multi-user preference studies were not completed, so the thesis does not convert informal demonstration feedback into numerical usability claims.

6.4 Integration results

The dynamic obstacle experiment provides the clearest integration evidence. A forward voice request remained active while an obstacle was physically introduced. Updated ultrasonic state caused the safety supervisor to stop the motor executor even though the voice source had not withdrawn its request. Removing the obstacle allowed the request to be reconsidered. This demonstrates continuous environmental reevaluation rather than a one-time check when speech was recognized.

The cold-boot experiment provides a second integration result: microphone discovery, speech model loading, ROS 2 nodes, sensor/motor configuration and local audio output can be brought into an operational state automatically after boot. Together these experiments establish that the voice component operates within the real embedded architecture rather than as an isolated desktop prototype.

6.5 Summary of findings

The principal objectives were achieved for core offline voice interaction and safety-aware motion integration. Local wake-word and movement commands worked on the Raspberry Pi; conversational memory and ML fallback were implemented; ROS 2 movement requests were physically connected to the robot through an independent safety layer; automatic startup was validated; and obstacle changes could override continued movement. Acoustic localization remains outside the validated result set.

7 DISCUSSION

7.1 Interpretation of findings

The results support the hypothesis that useful core voice interaction can be implemented on affordable embedded hardware without continuous cloud speech services. More importantly, the experiments show that speech recognition does not need to be trusted as a motor-authority mechanism. The robot can accept a valid command while still allowing lower-level environmental evidence to interrupt or block actuation.

This design reduces one class of embodied-AI risk: a high-level recognition or conversation component cannot unilaterally continue physical motion when the safety layer determines that the current environment is unsafe or stale. The architecture therefore treats language as intention and sensors as part of authorization.

7.2 Comparison with literature baselines and existing solutions

Large general-purpose ASR models can offer broader recognition capability [3], but the BUDDY implementation prioritizes local availability and a small deterministic command set for physical control. This is a deliberate engineering trade-off. In contrast to a cloud-first assistant, BUDDY can retain core wake, movement and local spoken-feedback functions without an internet connection. In contrast to a direct scripted motor interface, it retains a separate safety supervisor.

The layered control principle is consistent with behaviour-based robotics concepts in which higher-level goals do not override lower-level protective behaviour [9]. BUDDY realizes this using ROS 2 topics and typed messages, enabling future command sources to share the same authority boundary.

7.3 Research contribution

The individual research contribution is a physically validated architecture for offline voice interaction in an embodied companion robot where safety-relevant commands are deterministic and remain subordinate to independent sensor-based supervision. The contribution includes the implemented audio pipeline, wake-state design,

movement-request interface, conversational separation, persistent local memory, local TTS, automatic deployment and integration evidence.

7.4 Product contribution

From a product perspective, the voice subsystem makes the prototype approachable without requiring a phone or terminal for everyday basic commands. Automatic startup further reduces the gap between a research script and a usable appliance-like robot. Persistent local profile information supports future personalization, while the modular interface makes it possible to combine speech with visual and navigation components without rewriting the core voice logic.

7.5 Limitations and threats to validity

The main validity limitation is the absence of a controlled multi-participant speech study. Wake range and movement-command observations were obtained in a practical development environment rather than a standardized acoustic chamber. Recognition performance may differ with accent, background noise, microphone orientation, speaker level and room reflections. The reported 0-3 m usability range should therefore be interpreted as prototype evidence, not a general device specification.

The single validated microphone limits spatial hearing. Two identical AB13X USB microphones exhibited unreliable simultaneous capture on the Raspberry Pi, preventing defensible acoustic-source localization results. Future direction-of-arrival research should use synchronized multi-channel hardware or an interface designed for coherent capture rather than independent USB devices.

The prototype also lacks wheel encoders and an IMU for closed-loop heading. Voice requests can command turns, but exact angular rotation is not claimed. Safety thresholds are configuration parameters and were not derived from a formal calibrated ranging study. Finally, the implemented architecture reduces risk but is not a certified functional-safety system.

7.6 Implications

The findings suggest that companion robots do not require their most complex intelligence to control every layer directly. A small local command vocabulary can coexist with richer conversation if authority boundaries are explicit. This pattern is especially relevant for educational, research and low-cost domestic robots where resource constraints encourage modular software and where cloud connectivity cannot always be assumed.

8 COMMERCIALISATION AND SUSTAINABILITY

8.1 Value proposition and target users

BUDDY's value proposition is an affordable, modifiable companion-robot platform with local interaction, safety-aware movement and extensible perception. Potential users include robotics students, educational institutions, research laboratories and households interested in a transparent prototype rather than a closed premium robotic pet. The offline voice component contributes privacy, resilience and low recurring service dependence for core functions.

8.2 Market and competitive position

The system should not be positioned as a direct replacement for mature commercial robots. Its competitive strength is openness, modularity and cost-conscious construction. Commodity components can be replaced individually, and the software stack can be inspected and modified. This makes the platform especially appropriate for research and education, where access to internal data flow and control boundaries is valuable.

8.3 Business or sustainability model

A sustainable model could combine an open research core with assembled educational kits, integration support, curriculum material and optional advanced modules. Local speech reduces dependence on paid cloud inference for basic use. Long-term sustainability also depends on standard connectors, documented interfaces and replaceable batteries/sensors rather than proprietary sealed assemblies.

8.4 Scalability intellectual property and compliance

Scaling from a research prototype to a consumer product would require mechanical redesign, EMC and electrical testing, secure update mechanisms, privacy controls, stronger identity/access management, formal hazard analysis and relevant regulatory assessment. Open-source dependencies and model licences must be reviewed before commercial distribution. The present thesis makes no claim of consumer certification or freedom to operate.

8.5 Contribution to the Sustainable Development Goals

The project is most closely aligned with SDG 9 (Industry, Innovation and Infrastructure) through low-cost robotics innovation and technical capacity building, and SDG 4 (Quality Education) through an accessible platform for learning embedded AI, ROS 2, sensing and human-robot interaction. Local processing and modular repairability may also support more resource-conscious computing practices, although a formal lifecycle assessment was not performed.

9 CONCLUSION AND FUTURE DIRECTIONS

9.1 Conclusion

This thesis presented the offline voice-intelligence and safety-aware system-integration contribution to BUDDY. The completed prototype demonstrates that a Raspberry Pi-class companion robot can provide local wake-word interaction, offline movement commands, spoken feedback, conversational memory and modular ROS 2 integration while maintaining a distinct environmental safety boundary before physical actuation.

Physical experiments established automatic startup, usable wake interaction across the tested 0-3 m range, forward/backward/left/right/stop wheel control, four-direction ultrasonic sensing, dynamic interruption of an active forward request and resumption after obstacle removal. These results support the central argument that natural-language interaction can be connected to embodied motion without granting conversational intelligence direct motor authority.

9.2 Achievement of objectives

Objective	Outcome
O1 Local wake/ASR/TTS	Achieved for the physical prototype using Vosk and Piper.
O2 Separate movement grammar from conversation	Achieved; movement phrases are deterministic and handled before general conversation.
O3 Structured ROS 2 movement requests	Achieved through MovementCommand on /movement/request.
O4 Independent safety authorization	Achieved and physically demonstrated with dynamic obstacle intervention.
O5 Persistent memory and ML fallback	Implemented at software level.

O6 Automatic embedded startup	Physically validated through systemd deployment.
O7 Experimental evaluation	Completed for functional voice/motion/safety behaviour; generalized speech accuracy remains future work.

9.3 Individual research and product contributions

The candidate contributed the principal auditory interaction implementation, the separation of safety-critical movement grammar from conversation, the ROS 2 voice-to-request interface, persistent user context, local speech synthesis and significant system integration required for the physical demonstration. The work also produced reproducible deployment artefacts and a clear evidence boundary between validated single-microphone interaction and experimental dual-microphone localization.

9.4 Future directions

Future work should conduct controlled multi-speaker evaluation across noise levels, accent variation and microphone distance; measure end-to-end recognition and response latency under full multimodal load; replace the independent dual-USB microphone experiment with synchronized multichannel acquisition; and investigate robust acoustic source localization. The conversational layer can be extended with better local language models provided that physical movement remains outside unconstrained generation.

Additional work should also examine privacy-preserving long-term personalization, encrypted remote interfaces, power/runtime profiling, hardware acceleration and long-duration reliability. Any future autonomous behaviour should continue to route physical movement through the same independent safety-supervision principle.

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GLOSSARY

Term	Meaning
ASR	Automatic Speech Recognition
HRI	Human–Robot Interaction
ROS 2	Robot Operating System 2
TTS	Text-to-Speech
Vosk	Offline speech-recognition toolkit used by BUDDY
Piper	Local neural text-to-speech engine
TF-IDF	Term Frequency–Inverse Document Frequency
GPIO	General-Purpose Input/Output
PWM	Pulse-Width Modulation
BMS	Battery Management System
Wake word	Short phrase used to move BUDDY from sleeping to active interaction
MovementCommand	Typed ROS 2 message representing movement intent rather than direct motor authority

APPENDICES

Appendix A Research instruments or extended results

A.1 Physical voice and safety test checklist

Check	Expected evidence
Cold boot	System starts without manual terminal launch and announces readiness.
Wake phrase	“Buddy” produces “Yes?” in the tested environment.
Forward/back/left/right/stop	Corresponding physical wheel behaviour observed through the safety path.
Obstacle intervention	Active forward motion stops when a front obstacle is introduced.
Obstacle removal	Still-active request is reconsidered after hazard removal.
Ultrasonic channels	FL, FR, RL and RR observations available.
Shutdown	Processes stop and hardware cleanup occurs without uncontrolled continued motion.

A.2 Repository artefacts relevant to the individual contribution include
buddy_core/voice/buddy_voice.py, buddy_core/voice/brain.py,
buddy_core/voice/user_profile.py, buddy_core/voice/emotion_response.py,
ml_brain.py, train_model.py, intents.json,
buddy_interfaces/msg/MovementCommand.msg, buddy_core/safety/supervisor.py,
deployment/buddy.service and scripts/start_buddy_demo.sh.

Appendix B Academic integrity and AI use disclosure

Generative AI systems, including ChatGPT, were used during the project as development-support tools for debugging explanations, restructuring code ideas, language refinement, diagram planning and preparation of this thesis draft. AI-generated suggestions were not accepted as experimental evidence. Hardware values, repository details and result claims were checked against the candidate's own implementation, project records and observed tests. The candidate retains responsibility for code submitted to the project, final analysis, citations and all claims presented for assessment.

Where AI-assisted code or text was used, it was reviewed and adapted to the project context. No fabricated participant results, speech-accuracy percentages, safety-certification claims or unperformed experiments are included. The thesis explicitly labels incomplete work such as dual-microphone acoustic localization as experimental or not validated.

Appendix C Hardware and integration summary

Component	Final / validated role
Raspberry Pi 4B 4 GB	Primary embedded computer running Ubuntu 24.04 and ROS 2 Jazzy
AB13X USB microphone	Validated single-microphone input; 48 kHz stereo converted to 16 kHz mono
USB audio output / speaker	Validated local speech output
2 x TB6612FNG	Four-motor actuation
4 x DC geared motors	Physical movement platform
4 x HC-SR04	Four-direction obstacle observations
BSS138 level converter	Ultrasonic echo level conversion
PCA9685 + 2 x MG90-class servos	Two-axis camera/head actuation; visual subsystem integration support

3S 18650 + BMS + fuse + switch + buck	Separate regulated motor/servo power domain
20,000 mAh USB-C power bank	Separate Raspberry Pi compute supply

Appendix D Individual contribution mapping

Area	Candidate contribution	Evidence
Offline auditory interaction	Wake word, local ASR pipeline, command/general recognition separation	buddy_voice.py; physical wake tests
Conversational intelligence	Deterministic response logic, ML fallback, memory/profile support	brain.py, ml_brain.py, user_profile.py
Safety-aware motion integration	MovementCommand requests routed through safety supervisor	ROS 2 topics; dynamic obstacle experiment
Deployment	Autostart script and systemd service	start_buddy_demo.sh; buddy.service; cold-boot test
System integration	Pi, audio, motor/sensor, power and ROS 2 integration support	Hardware records and physical demo
Cross-component support	Assisted final integration/troubleshooting of visual/head system without claiming visual research ownership	Integration records; team contribution boundary

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